

TASK-EFFICIENT AGENT EVALUATION USING CACHED TRACES

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ABSTRACT

Efficient benchmarking estimates a system’s full-benchmark performance from a small number of task executions. Standard approaches rely solely on the binary correctness of executions from a collection of already-evaluated reference systems. An agentic system, however, produces a rich trace for each execution that can reveal substantially more about its capabilities. Exploiting this information requires useful trace representations. We propose representing a trace by extracting a structured, rubric-guided summary of it with an LLM and embedding the summary with a standard text encoder. These representations preserve correctness- and task-related signal while minimizing the presence of system identity-, model family-, and harness-related signals that dominate raw trace embeddings. Given this representation for each task execution from all reference systems, we construct low-dimensional representations of the systems, predict a new system’s score from its position among the reference systems, and combine the resulting geometric prediction with a two-parameter logistic item-response model. On SWE-bench Verified and Terminal-Bench 2.0, the proposed estimator reduces mean absolute error from 0.13 to 0.08 ($p < 10^{-4}$) and from 0.14 to 0.12 ($p < 10^{-3}$), respectively, when provided a single random executed task relative to an item-response model alone. These results indicate that trace geometry and correctness outcomes carry complementary predictive signal and enable more accurate benchmark score estimation from limited agent executions.

1 INTRODUCTION

The dominant paradigm of generative model use is increasingly agentic: systems plan, invoke tools, observe intermediate results, and iterate over long horizons to complete complex tasks (Yang et al., 2024; Wang et al., 2025). As agentic capabilities increase, so does the range of settings in which a given system might be deployed – and, with it, the number of benchmarks on which the system must be evaluated before it can be trusted with consequential work (Kapoor et al., 2026). Agentic evaluation, however, is expensive as each task execution requires a properly sandboxed environment and a single task execution routinely consumes hundreds of thousands of tokens (Jimenez et al., 2024; Terminal-Bench Team, 2026) and can regularly exceed \$2,000 (Kapoor et al., 2026). Given this cost, efficient evaluation is necessary to increase the evaluation coverage of an agentic system.

Task-efficient evaluation methods (e.g., benchmark compression and anchor-item selection (Vivek et al., 2024; Maia Polo et al., 2024), item response theory and adaptive testing (Kipnis et al., 2025; Truong et al., 2025), leveraging cached responses (Helm et al., 2026a), etc.) address this problem by finding structure in item-level information available from previously-evaluated reference systems to decrease the cost of evaluating a new system. Some task-efficient evaluation methods require only the item-level scores from reference systems (Vivek et al., 2024; Kipnis et al., 2025). Others leverage the content of cached artifacts, such as natural language responses or residual streams, to measure the similarity of systems on a set of tasks (Duderstadt et al., 2023; Helm et al., 2026b;a). Notably, agentic workflows often produce artifacts when executing a task – file diffs, tool calls, intermediate plans, and error messages – recorded together in an execution trace that can be used to this effect. Unfortunately, existing methods for using cached artifacts rely on off-the-shelf embedding functions that are not immediately well-suited for representing traces.

We make two primary contributions:

Table 1: Mean Absolute Error (MAE) of benchmark score estimators for SWE-bench Verified and Terminal-Bench 2.0 under leave-one-family-out evaluation and different task selection protocols (random & adaptive). **Bolded** values are the best method for a given benchmark, task selection, and number of tasks. Underlined values are statistically significant ($\alpha = 0.05$) comparisons to IRT per the two-sided Wilcoxon signed-rank test. The TRUBRIC and IRT blend – which takes advantage of both trace geometry and per-execution score – is the best or near-best in all settings.

Num. tasks m	SWE-bench Verified						Terminal-Bench 2.0					
	random			adaptive			random			adaptive		
	1	5	20	1	5	20	1	5	20	1	5	20
sample score	0.442	0.175	0.082	0.380	0.151	0.072	0.396	0.164	0.073	0.360	0.124	0.132
IRT	0.127	0.088	0.048	0.080	0.062	0.034	0.137	0.097	0.052	0.112	0.067	0.039
raw trace	<u>0.099</u>	0.093	0.089	0.096	0.083	0.081	0.135	0.126	0.120	0.128	0.125	0.113
generic	0.079	<u>0.071</u>	0.068	0.074	0.070	0.069	0.139	0.127	0.115	0.144	0.130	0.117
trubric	<u>0.081</u>	<u>0.069</u>	0.060	0.068	0.056	0.060	<u>0.127</u>	0.107	0.093	0.110	0.092	0.084
IRT + trubric	<u>0.081</u>	0.063	0.043	0.062	0.048	0.034	0.122	0.090	0.050	0.099	0.059	0.037

1. **TRUBRIC trace representations.** We introduce TRUBRIC (*trace rubric*): a trace representation constructed by extracting a task specific structured summary of each trace with an LLM and embedding the summary with a standard text encoder. We show that TRUBRIC embeddings preserve information relevant to the task and behavior of the system while minimizing system-, model family-, and harness- specific signal across seven modern text encoders.

2. **Task-efficient evaluation using cached traces.** Given the TRUBRIC embeddings from a set of tasks for a target system and a collection of reference systems, we evaluate their utility for predicting the target system’s full-benchmark score on SWE-bench Verified and Terminal-Bench 2.0 under three different agentic system evaluation protocols. We find that TRUBRIC-based estimators, combined with an item-response model, reduce single-task mean absolute error from 0.13 to 0.08 ($p < 10^{-4}$) and from 0.14 to 0.12 ($p < 10^{-3}$) on the two benchmarks, respectively, and are nearly always better than baselines at any task budget (Table 1).

1.1 PROBLEM STATEMENT

For our purposes, an agentic system $f \in \mathcal{F}$ is a random mapping from a task space $\mathcal{Q}$ to a trace space $\mathcal{X}$. A benchmark is a fixed collection of tasks $Q^* = \{q_1, \dots, q_M\} \subset \mathcal{Q}$. Each trace is evaluated by the benchmark’s trace-level scoring function $s : \mathcal{X} \rightarrow \{0, 1\}$ and the system is evaluated by its benchmark score $y(f) = \frac{1}{M} \sum_{j=1}^M \mathbb{E}[s(f(q_j))]$ where the expectation is with respect to the randomness of f . We write y for $y(f)$ when the context is clear.

When evaluating a new system, we typically have access to $\mathcal{D} = \{\{(x_{ij}, y_{ij})\}_{j=1}^M\}_{i=1}^n$, the traces and trace-level scores of n previously-evaluated reference systems $f_1, \dots, f_n$, where $x_{ij} \sim f_i(q_j)$, $y_{ij} = s(x_{ij})$ and $y_i = y(f_i)$. For the target system f_0 , we observe traces (and, optionally, trace-level scores) on only a subset $Q \subset Q^*$ of $|Q| = m \ll M$ tasks and our goal is to estimate the system’s full benchmark score y_0 using only the target’s m executions and $\mathcal{D}$.

2 METHOD

3 TRACE REPRESENTATIONS ENCODE CONFOUNDERS

4 EFFICIENT EVALUATION

5 RELATED WORK

Representing traces and structured artifacts. The standard tools for embedding model outputs are text encoders trained on sentence-level semantic similarity (Reimers & Gurevych, 2019; Su et al., 2023; Nussbaum & Duderstadt, 2025). Agentic traces, however, implicitly or explicitly encode the

scaffold or underlying model that may not be relevant for downstream tasks (Sun et al., 2025; Oderinwale, 2026) and are not necessarily suitable to be encoded. Instead, traces have primarily been studied through taxonomies of failure modes and actions (Cemri et al., 2025; Shu et al., 2026), LLM judges of agent behavior (Zhuge et al., 2025; Mehtiyev & Assunção, 2026), and trajectory compression for cost reduction (Xiao et al., 2025). Further, parallel work adapts model outputs to better fit the notion of semantic similarity either by comparing structured texts through natural-language descriptions of their properties (Ravfogel et al., 2024) or by representing an output by an LLM’s answers to a set of questions about it (Peng et al., 2024; Benara et al., 2024). TRUBRIC applies this to agentic traces by embedding a task-specific structured summary of each trace, preserving behavior-relevant information while minimizing system-level information.

Efficient model evaluation. Task-efficient evaluation methods estimate a system’s full-benchmark score from a small number of executions. Item response theory (IRT) is the workhorse of score-only approaches: each system is assigned a latent ability, each task a difficulty and discrimination, and a logistic link models the probability of success, so that an ability estimated from m graded executions yields a predicted benchmark score (Maia Polo et al., 2024; Kipnis et al., 2025). Extensions select informative or adaptive task subsets (Vivek et al., 2024; Truong et al., 2025) and apply IRT to agentic benchmarks directly (Ge et al., 2026; Ndzomga, 2026), though the reliability of small-budget estimates remains contested (Yauney et al., 2026). A complementary family represents the systems themselves, either from their score profiles (Ruan et al., 2024; Zhuang et al., 2025) or from the content of their cached responses. The Data Kernel Perspective Space (DKPS) embeds each system’s responses to a common set of tasks and represents systems as points obtained by classical multidimensional scaling of the distances between their embedded responses (Duderstadt et al., 2023; Helm et al., 2025; Acharyya et al., 2024). Most relevant to our setting, Helm et al. (2026a) regress benchmark scores on DKPS representations and show that the resulting predictions are task-efficient relative to the sample score. TRUBRIC extends this geometry to agentic traces and combines it with a two-parameter IRT model, exploiting the complementary signal in traces and scores.

6 DISCUSSION

REFERENCES

- Aranyak Acharyya, Michael W. Trosset, Carey E. Priebe, and Hayden S. Helm. Consistent estimation of generative model representations in the data kernel perspective space. *arXiv preprint arXiv:2409.17308*, 2024.
- Vinamra Benara et al. Crafting interpretable embeddings by asking LLMs questions. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2405.16714.
- Mert Cemri et al. Why do multi-agent LLM systems fail? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2503.13657.
- Brandon Duderstadt, Hayden S. Helm, and Carey E. Priebe. Comparing foundation models using data kernels. *arXiv preprint arXiv:2305.05126*, 2023.
- Yuan Ge et al. Agent psychometrics: Item response theory for agentic coding evaluation. *arXiv preprint arXiv:2604.00594*, 2026.
- Hayden Helm, Brandon Duderstadt, Youngser Park, and Carey Priebe. Statistical inference on black-box generative models in the data kernel perspective space. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. arXiv:2410.01106.
- Hayden Helm, Johnson, and Carey Priebe. Query-efficient model evaluation using cached responses. In *International Conference on Machine Learning (ICML)*, 2026a. arXiv:2605.07096.
- Hayden Helm, Xiaodong Liu, and Weiwei Yang. Jailbreak susceptibility prediction and mitigation via the behavioral geometry of models. *arXiv preprint arXiv:2605.26409*, 2026b.

- Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. SWE-bench: Can language models resolve real-world github issues? In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.06770.
- Sayash Kapoor, Benedikt Stroebel, et al. Holistic agent leaderboard: The missing infrastructure for AI agent evaluation. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2510.11977.
- Alex Kipnis et al. metabench: A sparse benchmark to measure general ability in large language models. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2407.12844.
- Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. tinyBenchmarks: Evaluating LLMs with fewer examples. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.14992.
- Rustam Mehtiyev and Wesley K. G. Assunção. Beyond resolution rates: Behavioral drivers of coding agent success and failure. *arXiv preprint arXiv:2604.02547*, 2026.
- Michel Ndzomga. Efficient benchmarking of AI agents. *arXiv preprint arXiv:2603.23749*, 2026.
- Zach Nussbaum and Brandon Duderstadt. Training sparse mixture of experts text embedding models. *arXiv preprint arXiv:2502.07972*, 2025.
- Hamidah Oderinwale. Agent trajectories as programs: Attribution of coding-agent traces to scaffold and model. *arXiv preprint arXiv:2606.16988*, 2026.
- Letian Peng et al. Answer is all you need: Instruction-following text embedding via answering the question. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. arXiv:2402.09642.
- Shauli Ravfogel, Yoav Goldberg, and Ryan Cotterell. Description-based text similarity. In *Conference on Language Modeling (COLM)*, 2024. arXiv:2305.12517.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019. arXiv:1908.10084.
- Yangjun Ruan, Chris J. Maddison, and Tatsunori Hashimoto. Observational scaling laws and the predictability of language model performance. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2405.10938.
- Wei Shu et al. What resolve rate hides: Canonical action taxonomies for coding agent trajectories. *arXiv preprint arXiv:2607.06184*, 2026.
- Hongjin Su et al. One embedder, any task: Instruction-finetuned text embeddings. In *Findings of the Association for Computational Linguistics (ACL)*, 2023. arXiv:2212.09741.
- Mingjie Sun et al. Idiosyncrasies in large language models. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.12150.
- Terminal-Bench Team. Terminal-bench: Benchmarking agents on hard, realistic tasks in command line interfaces. *arXiv preprint arXiv:2601.11868*, 2026.
- Sang Truong et al. Reliable and efficient amortized model-based evaluation. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2503.13335.
- Rajan Vivek, Kawin Ethayarajh, Diyi Yang, and Douwe Kiela. Anchor points: Benchmarking models with much fewer examples. In *Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2024. arXiv:2309.08638.
- Xingyao Wang et al. OpenHands: An open platform for AI software developers as generalist agents. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2407.16741.
- Yiran Xiao et al. AgentDiet: Reducing the cost of LLM agents with trajectory reduction. *arXiv preprint arXiv:2509.23586*, 2025.

John Yang, Carlos E. Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2405.15793.

Gregory Yauney et al. How reliable is language model micro-benchmarking? In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2510.08730.

Richard Zhuang et al. EmbedLLM: Learning compact representations of large language models. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.02223.

Mingchen Zhuge et al. Agent-as-a-judge: Evaluating agents with agents. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2410.10934.